

HMF Design Clinic Grading Guide

A case study score, an overall score, and qualitative feedback on three dimensions

Score each case study across its four pain points, out of 100, for an overall total of 400. Alongside the score, assess each team on three dimensions and record qualitative feedback. Use the Solution Guide for the model answers.

The three dimensions

Problem Identifier Correctly identified the customer's pain point and the requirements behind it.	Solution Designer Recommended the right Cisco solution, architecture, and approach.	Value Communicator Clearly explained how the solution works and the customer outcomes it enables.
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How it works. For each case study, read the team's four answers, assess them on the three dimensions using the scale below, and assign a **case study score out of 100**. Add the four case study scores for an **overall score out of 400**. Grade against the Solution Guide and reward reasoning specific to Cedarline.

The assessment scale

Strong	Consistently meets the dimension across the case study; specific to Cedarline.
Developing	Solid, but with gaps or generic reasoning in places.
Emerging	Early. The idea is there but thin, partial, or unclear.

From ratings to a score

Use the overall picture across the three dimensions to place the case study score in a range.

Mostly Strong	85 to 100 / 100
Mostly Developing	65 to 84 / 100
Mostly Emerging	40 to 64 / 100
Below Emerging	under 40 / 100

Evaluating a team's submission

Each team submits a completed Workbook. Work through it one case study at a time using the broken-down expected answers on the pages that follow.

- 1 Collect each team's Workbook and check that all four case studies have been attempted.
- 2 Take one case study at a time. Read the team's four answers alongside the broken-down expected answers on this page.
- 3 For each pain point, compare the team's Product / platform and Solution Engineer response against the expected Problem, Solution and Value.
- 4 Assess the case study on the three dimensions using the Strong, Developing, Emerging scale.
- 5 Use the ratings-to-score mapping to place the case study score out of 100, then write one line of strengths and coaching.
- 6 Add the four case study scores for the overall score out of 400, and pass the topology diagram to the SME for review.

What each dimension looks for. **Problem Identifier:** did the team name the real requirement behind the concern? **Solution Designer:** did they land on the right product and an approach that fits Cedarline? **Value Communicator:** did they explain how it works and the outcome? Compare against the expected answers, and reward reasoning specific to Cedarline.

1

Case Study 1

Compare each team answer against the expected answer, then assess the case study below.

PAIN POINT 1 Expected answer

Under the EU AI Act, we are classified as a 'Provider' for our internal AI systems. We need to demonstrate robustness and cybersecurity compliance to satisfy Article 15. How does Cisco help us meet these specific requirements?

Problem	Provider-level evidence of robustness and cybersecurity under EU AI Act Article 15.
Solution	Cisco AI Defense (AI Model and Application Validation)
Value	How it works. Cisco AI Defense is specifically designed to support Provider obligations under the EU AI Act. We utilize algorithmic red teaming technology for AI Model & App Validation, which identifies safety and security vulnerabilities at scale. Outcome. It automates the technical evidence gathering required for "Conformity Assessments." By mapping AI-specific threats to standardized frameworks (like MITRE ATLAS and OWASP), it provides the documentation and continuous monitoring necessary to prove to regulators that your systems are robust and compliant.

PAIN POINT 2 Expected answer

We are integrating various open-source models and third-party agents into our internal workflows. How can we ensure these assets aren't introducing vulnerabilities into our environment before they even go live?

Problem	Vetting AI models and third-party agents for vulnerabilities before they reach production.
Solution	Cisco AI Defense (Supply Chain Risk Management)
Value	How it works. Cisco AI Defense addresses this through proactive Supply Chain Risk Management. We scan your model files, repositories, and MCP servers to identify malicious or unsafe assets. Outcome. It enables "Shift-Left" security by creating an inventory of all AI models and agents. By identifying vulnerabilities before production, it prevents the introduction of malicious code or poisoned data into your ecosystem.

PAIN POINT 3

Expected answer

We know AI is showing up across our cloud environments faster than security can track it. We're worried we have models, agents, datasets, and MCP-connected tools running without visibility or consistent controls.

Problem	Visibility and consistent control over AI assets (models, agents, datasets and MCP tools) across the cloud.
Solution	Cisco AI Defense (AI Cloud Visibility)
Value	How it works. That's exactly the visibility gap Cisco AI Defense is built to solve. With AI Cloud Visibility, Cisco AI Defense continuously discovers AI assets across distributed cloud environments, including models, agents, datasets, MCP servers, and agent-to-tool workflows. Outcome. Identifies rogue or unsanctioned AI assets. Maps relationships across data, models, agents, and tools.

PAIN POINT 4

Expected answer

We're running AI workloads in cloud accounts, but we don't have a reliable way to see which models and AI assets are actually in use. We're also concerned about workloads reaching out to third-party AI models without security knowing about it. And even when we do know the traffic path, we need a practical way to inspect and enforce guardrails on prompt and response traffic in the cloud without changing every application.

Problem	Seeing AI assets and third-party model usage, and enforcing runtime guardrails on AI traffic without changing applications.
Solution	Cisco AI Defense with Cisco Multicloud Defense (AI Runtime Protection)
Value	How it works. That's exactly what the Cisco AI Defense-Cisco Multicloud Defense integration is designed to address. Cisco AI Defense uses the integration to discover AI assets in your cloud environments and identify external AI destinations your workloads are calling. Outcome. Finds AI assets in cloud accounts. Identifies third-party AI model usage.

ASSESS CASE STUDY 1

DIMENSION	ASSESSMENT (CIRCLE ONE)	NOTES
Problem Identifier	Strong Developing Emerging	
Solution Designer	Strong Developing Emerging	
Value Communicator	Strong Developing Emerging	

Case Study 1 score _____ / 100

Strengths and coaching for this case study

2

Case Study 2

Compare each team answer against the expected answer, then assess the case study below.

PAIN POINT 1 Expected answer

If a breach occurs in one of our public cloud instances, we are terrified that the attacker will move laterally into our core on-premises systems. How can we contain these threats in a multicloud environment?

Problem	Containing lateral movement between public cloud and on-prem in a multicloud environment.
Solution	Cisco Multicloud Defense (macro and micro-segmentation gateways)
Value	How it works. Cisco Multicloud Defense leverages macro and micro-segmentation capabilities to enforce strict communication boundaries. By integrating with our Hybrid Mesh Firewall and Hypershield, we can isolate workloads and inspect traffic between cloud segments, effectively stopping lateral movement before it reaches your sensitive data. Outcome. It provides granular visibility and control over east-west traffic. By applying Zero Trust principles across the hybrid fabric, you transform your network into a series of secure, isolated zones that prevent an attacker from pivoting through your infrastructure.

PAIN POINT 2 Expected answer

Our auditors require proof that our security controls are applied consistently across all our cloud workloads. Gathering this data from different cloud providers is a manual, time-consuming nightmare.

Problem	Consistent, auditable proof that security controls are applied across all cloud workloads.
Solution	Cisco Multicloud Defense (unified logging and visibility)
Value	How it works. Cisco Multicloud Defense provides unified logging and traceability. Because our security framework is integrated across the network fabric, you can generate consolidated reports that map directly to your compliance requirements (such as NIS2 or DORA) across all cloud environments from a single dashboard. Outcome. It automates the collection of evidence for security audits. By consolidating logs and policy enforcement data, you create a "single source of truth" that simplifies compliance reporting and demonstrates proactive governance to regulators.

PAIN POINT 3

Expected answer

We have workloads spread across AWS, Azure, and on-premises data centers. Managing security policies individually in each cloud environment is creating massive operational overhead and leading to inconsistent security posture.

Problem	One consistent policy model across AWS, Azure and on-prem, without per-cloud overhead.
Solution	Cisco Multicloud Defense (unified policy across public clouds)
Value	How it works. Cisco Multicloud Defense, integrated within our Hybrid Mesh Firewall architecture, provides a unified control plane. You can define security policies once via Security Cloud Control and enforce them consistently across your entire distributed environment, regardless of the underlying cloud provider. Outcome. It replaces siloed, manual configuration with centralized orchestration. By abstracting the security policy from the cloud-native tools, it ensures that your security posture is uniform and eliminates the "configuration drift" that occurs when managing multiple cloud consoles.

PAIN POINT 4

Expected answer

We have workloads on-prem, in cloud, and in mixed environments, and policy consistency is becoming difficult. Every environment feels different, and we're worried about drift.

Problem	Unified segmentation that prevents policy drift across hybrid and multicloud.
Solution	Cisco Secure Workload with Cisco Secure Firewall (managed via FMC and Security Cloud Control)
Value	How it works. That's a common hybrid environment challenge. Cisco Secure Workload helps create a more unified segmentation model by using application dependency awareness and AI/ML-driven policy discovery to define policy based on how workloads actually communicate. Outcome. Reduces policy drift across hybrid and multicloud environments. Aligns policy to application behavior, not just static IP-based rules.

ASSESS CASE STUDY 2

DIMENSION	ASSESSMENT (CIRCLE ONE)	NOTES
Problem Identifier	Strong Developing Emerging	
Solution Designer	Strong Developing Emerging	

DIMENSION	ASSESSMENT (CIRCLE ONE)	NOTES
Value Communicator	Strong Developing Emerging	

Case Study 2 score _____ / 100

Strengths and coaching for this case study

3

Case Study 3

Compare each team answer against the expected answer, then assess the case study below.

PAIN POINT 1

Expected answer

When our security tools detect a compromised device, our manual response time is too slow. By the time we identify the user and manually isolate the port, the attacker has already moved through the network.

Problem	Automated, real-time isolation of a compromised device, faster than a manual response.
Solution	Cisco ISE (Security Group Tags via pxGrid and Adaptive Network Control)
Value	How it works. Cisco ISE integrates with your security ecosystem via pxGrid and Adaptive Network Control (ANC). When a threat is detected, ISE can automatically trigger a quarantine policy, isolating the compromised device in real-time without requiring manual intervention. Outcome. It automates the "Response" phase of the incident lifecycle. By bridging the gap between detection and enforcement, it stops threats at the speed of the network.

PAIN POINT 2

Expected answer

Our firewall environment has become too complex to manage manually.

Problem	Simplifying and cleaning up firewall policy administration at scale.
Solution	Cisco Security Cloud Control with FMC and Cisco Secure Firewall (unified policy administration)
Value	How it works. Cisco applies AI-driven analysis to firewall telemetry and policy data to surface optimization opportunities, reduce manual troubleshooting, and guide cleaner policy administration. Outcome. Reduces manual policy review; identifies redundant/stale rules; lowers risk from human error.

Most of our traffic is encrypted, so we've lost visibility into a huge part of our environment.

Problem	Restoring visibility into encrypted traffic without decrypting everything.
Solution	Cisco Secure Firewall with the Encrypted Visibility Engine (EVE)
Value	How it works. Encrypted Visibility Engine (EVE) uses AI/ML-based fingerprinting to analyze encrypted traffic patterns and identify suspicious behavior without decrypting the payload. Outcome. Restores visibility into encrypted blind spots; avoids the performance/privacy overhead of decrypt-everything strategies.

We're worried traditional signature-based detection won't catch new or unknown attacks quickly enough.

Problem	Detecting unknown and zero-day threats beyond signature-based detection.
Solution	Cisco Secure Firewall with Snort ML
Value	How it works. That's exactly where Snort ML helps. It adds machine-learning-based detection inside the Snort 3 engine to identify exploit behavior associated with unknown/zero-day threats. Outcome. Addresses the gap between known and unknown threat detection; reduces dependence on signature timing.

ASSESS CASE STUDY 3

DIMENSION	ASSESSMENT (CIRCLE ONE)	NOTES
Problem Identifier	☐ Strong ☐ Developing ☐ Emerging	
Solution Designer	☐ Strong ☐ Developing ☐ Emerging	
Value Communicator	☐ Strong ☐ Developing ☐ Emerging	

Case Study 3 score _____ / 100

Strengths and coaching for this case study

4

Case Study 4

Compare each team answer against the expected answer, then assess the case study below.

PAIN POINT 1 Expected answer

Network policy helps, but we also need to know what's happening at runtime inside the workload. We need better detection of suspicious behavior and policy violations.

Problem	Runtime detection of suspicious behaviour and policy violations inside workloads.
Solution	Isovalent Tetragon (eBPF runtime security)
Value	How it works. That's where Tetragon comes in. Tetragon extends the value of Cilium by providing eBPF-based runtime security and threat detection. Outcome. Adds runtime visibility beyond basic network controls. Detects suspicious system and network behavior in real time.

PAIN POINT 2 Expected answer

Our Kubernetes networking feels fragmented across clusters and environments. We're dealing with too many moving parts, inconsistent policies, and too much operational complexity.

Problem	Consistent Kubernetes networking and policy across clusters, with less operational complexity.
Solution	Isovalent Cilium (eBPF Kubernetes networking and policy)
Value	How it works. That's exactly where Isovalent Cilium helps. Cilium provides a modern, eBPF-based Kubernetes networking layer that gives you consistent connectivity, policy enforcement, and visibility across clusters and environments. Outcome. Replaces brittle, IP-centric networking models with a Kubernetes-native approach. Improves consistency across dynamic cluster environments.

Right now we're stitching together separate tools for networking, security, and observability in Kubernetes, and it's creating too much overhead.

Problem	Consolidating Kubernetes networking, security and observability to reduce tool sprawl.
Solution	Isovalent Cilium with Tetragon
Value	How it works. That's exactly the consolidation opportunity Isovalent addresses. With Cilium for networking and policy, and Tetragon for runtime security, you get a more unified platform for connectivity, segmentation, observability, and security. Outcome. Consolidates multiple functions into one platform approach. Reduces tool sprawl and operational friction.

We know we need microsegmentation, but we don't have enough visibility into east-west flows or application dependencies to create policy safely. We don't want to break the application.

Problem	Mapping east-west flows and application dependencies to segment safely without breaking applications.
Solution	Cisco Secure Workload (application dependency mapping)
Value	How it works. That's one of the biggest reasons customers adopt Cisco Secure Workload. It gives you deep visibility into workload communication and application dependency mapping, so you can see how applications actually behave before enforcing policy. Outcome. Reveals actual application communication paths. Reduces the risk of segmentation causing outages.

ASSESS CASE STUDY 4

DIMENSION	ASSESSMENT (CIRCLE ONE)	NOTES
Problem Identifier	☐ Strong ☐ Developing ☐ Emerging	
Solution Designer	☐ Strong ☐ Developing ☐ Emerging	
Value Communicator	☐ Strong ☐ Developing ☐ Emerging	

Case Study 4 score _____ / 100

Strengths and coaching for this case study

Team total

Team _____ Event _____

CASE STUDY	SCORE / 100
Case Study 1	_____
Case Study 2	_____
Case Study 3	_____
Case Study 4	_____
Overall total	_____ / 400

Topology diagram: SME review. The team's Hybrid Mesh Firewall diagram is submitted for Subject Matter Expert review and is not part of the score. Reviewed by SME: Yes ☐ No ☐

Overall qualitative feedback

Strengths across the clinic

Priorities to develop
